CS540 Introduction to Artificial Intelligence
Deep Learning I: Convolutional Neural Networks

University of Wisconsin-Madison
Spring 2025

Announcements

- **Homeworks:**
  - HW6 online, deadline on Monday **March. 17th at 11:59 PM**

- Class roadmap:

Machine Learning: Deep Learning I	Deep Learning
Machine Learning: Deep Learning II	
Machine Learning: Deep Learning II	

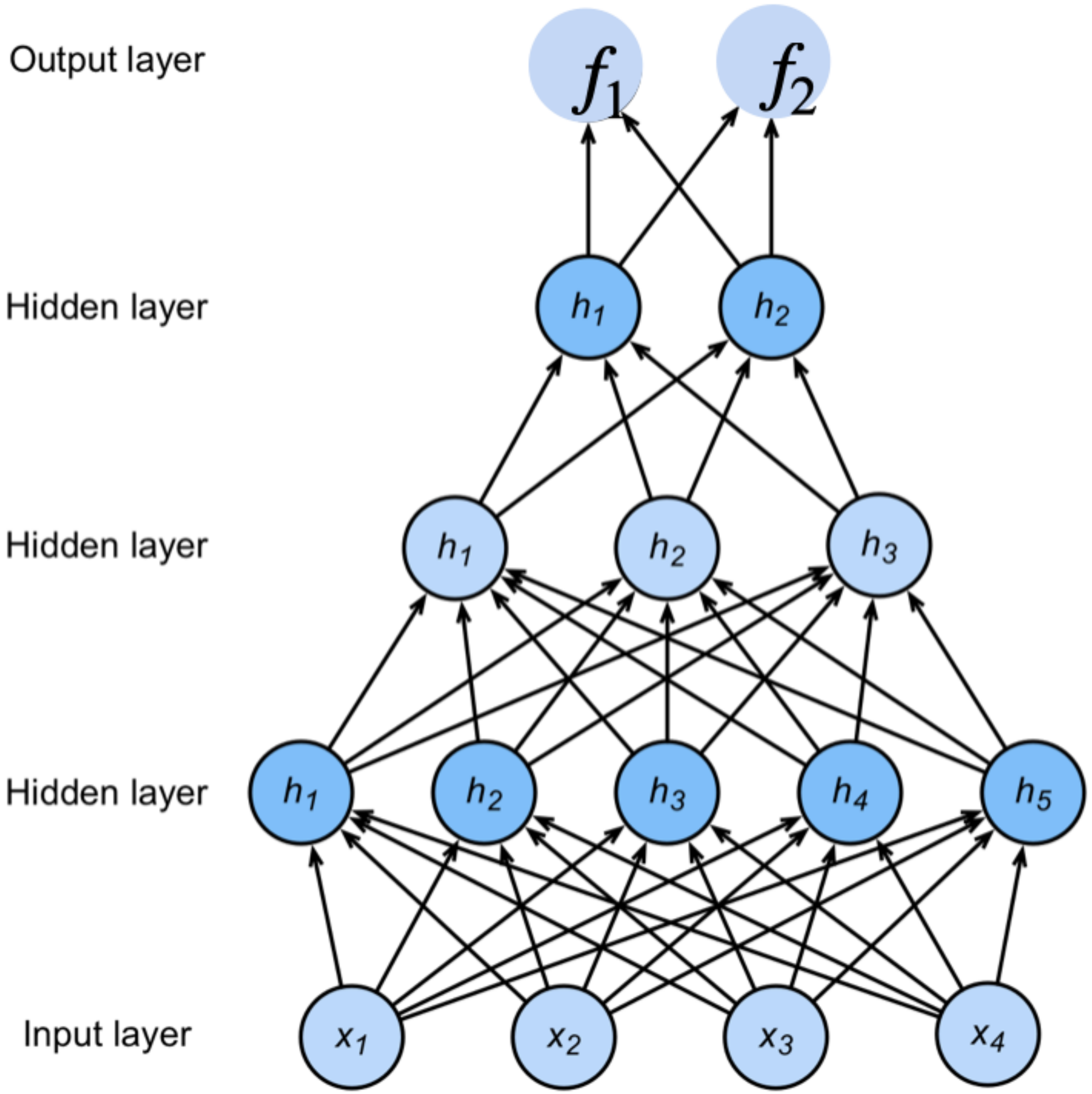
Midterm Information

- **Time:** March 13th 7:30-9 PM
- **Location:**
  - Section 001 : Ingraham Hall B10
  - Section 002 : Psychology 105
  - **Section 003: split in two locations according to the last name:**
    - **Chamberlin Hall 2103 (last name starting with A-L)**
    - **Sterling Hall 1310 (last name starting with M-Z)**
- McBurney students and students requesting alternate: reach out to your instructor if you have not received any email!
- Format: multiple choice
- WISC ID
- Cheat sheet: single piece of paper, front and back
- Calculator: fine if it doesn't have an Internet connection
- Detailed topic list + practice on Piazza and Canvas

Today’s Goals

- Build an understanding of convolutional neural networks.
- Why do we want convolutional layers?
- What are convolutional neural networks?
  - 2D vs 3D convolutional networks.
  - Padding and stride.
  - Multiple input and output channels
  - Pooling

Review: Multi-Layer Neural Networks



$$\mathbf{h}_1 = \sigma(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$$

$$\mathbf{h}_2 = \sigma(\mathbf{W}^{(2)}\mathbf{h}_1 + \mathbf{b}^{(2)})$$

$$\mathbf{h}_3 = \sigma(\mathbf{W}^{(3)}\mathbf{h}_2 + \mathbf{b}^{(3)})$$

$$\mathbf{f} = \mathbf{W}^{(4)}\mathbf{h}_3 + \mathbf{b}^{(4)}$$

$$\mathbf{p} = \text{softmax}(\mathbf{f})$$

NNs are composition of nonlinear functions

(★ Motivation here is that we would like a new kind of NN for images. For the example below, an image often involves lots of pixels, in this case 36M. That will be the dimension of our feature vector. Then you can imagine how big the NN we need to train such a model. Even with only one hidden layer of 100 neurons, that will be 36M x 100 weight parameters!)

How to classify

Cats vs. dogs?

Dual

12MP

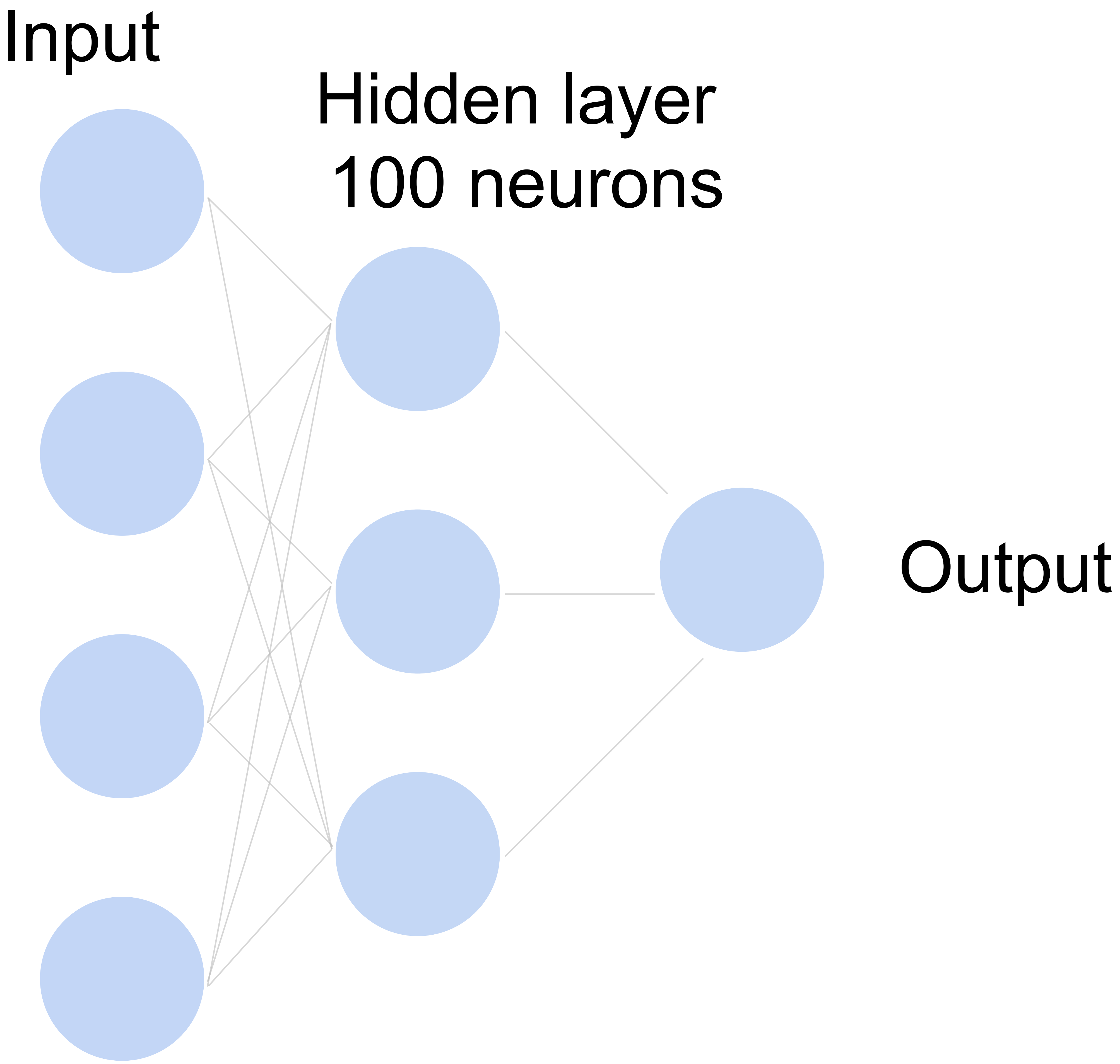
wide-angle and telephoto cameras

36M floats in a RGB image!

Fully Connected Networks

(Same with multi-layer perceptrons)

Cats vs. dogs?



~ 36M elements x 100 = **~3.6B** parameters!

Convolutions come to rescue!

Where is Waldo?

Why Convolution?

- Translation Invariance
- Locality

For Translation Invariance. We sometimes may move the image a little bit upwards and visually we know there is not much difference compared to the original one. However, the feature vectors will be completely different for the NN. That's why we need to track of it

For locality, we can often look at the local region of an image and that will often say something of the whole image.

⇓

(eg In general, there may not be a high correlation between a pixel at the top-right corner with another pixel at the other edge)

(☆ create a patch and move it step by step)

2-D Convolution

(★ The input itself does not change, the kernel slides over different patches of the input. Each patch is a submatrix of the input and has same dimensions as the kernel)

Input

This is the sliding patch

0	1	2
3	4	5
6	7	8

*

Filter

Kernel

0	1
2	3

(Dot Product)

=

Output

19	25
37	43

$$0 \times 0 + 1 \times 1 + 3 \times 2 + 4 \times 3 = 19$$

2-D Convolution

Input

0	1	2
3	4	5
6	7	8

*

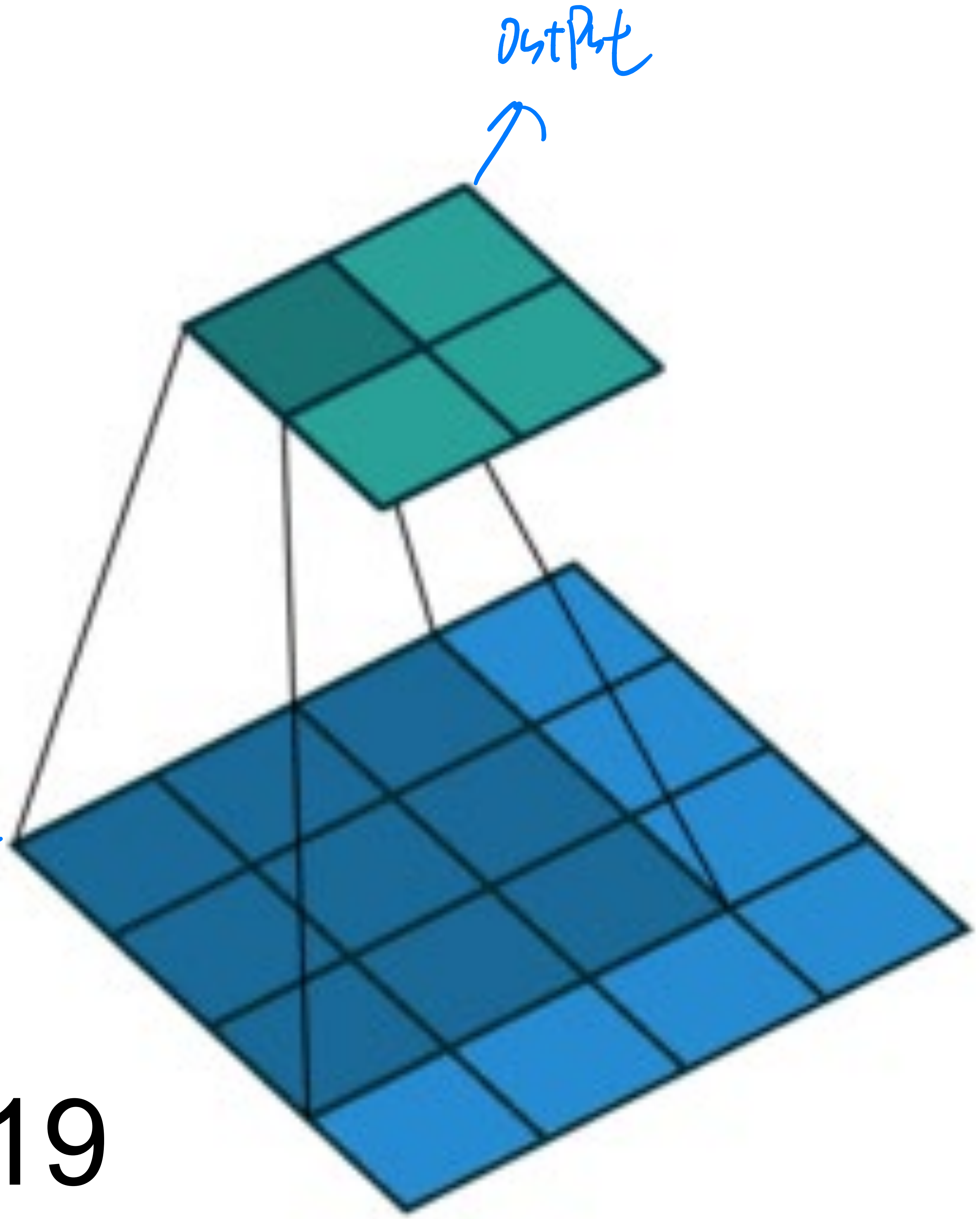
Kernel

0	1
2	3

=

Output

19	25
37	43



☆ 3x3 regions are patches
4x4 region is the whole input

$$0\text{x}0 + 1\text{x}1 + 3\text{x}2 + 4\text{x}3 = 19$$

(vdumoulin@ Github)

2-D Convolution

Input

→ we move left

0	1	2
3	4	5
6	7	8

*

Kernel

0	1
2	3

=

Output

19	25
37	43

$$1 \times 0 + 2 \times 1 + 4 \times 2 + 5 \times 3 = 25$$

2-D Convolution

Input | Kernel | Output

move down

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 19 & 25 \\ 37 & 43 \end{bmatrix}$$

$$3x0 + 4x1 + 6x2 + 7x3 = 37$$

2-D Convolution

Input — the last one

Kernel

Output — Output matrix

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 19 & 25 \\ 37 & 43 \end{bmatrix}$$

$$4x0 + 5x1 + 7x2 + 8x3 = 43$$

2-D Convolution Layer

0	1	2
3	4	5
6	7	8

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 19 & 25 \\ 37 & 43 \end{bmatrix}$$

height width

- **X**: n_h x n_w input matrix
- **W**: k_h x k_w kernel matrix
- **Y**: $(n_h - k_h + 1) \times (n_w - k_w + 1)$ output matrix

☆ Number of places in height that our patch can fit

(e.g. 3-2+1= 2 → 2x2 in output)

☆ Convolution operator not multiplication

Y = X * W

2-D Convolution Layer

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} + 1 = \begin{bmatrix} 20 & 26 \\ 38 & 44 \end{bmatrix}$$

- **X**: n_h x n_w input matrix
- **W**: k_h x k_w kernel matrix
- b: scalar bias → Just add them into the output
- **Y**: $(n_h - k_h + 1) \times (n_w - k_w + 1)$ output matrix

convolution! Not multiplication!

Y = X * W + b

- **W** and *b* are learnable parameters

Examples

(wikipedia)

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \rightarrow$$

Edge Detection

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} \rightarrow$$

Sharpen

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \rightarrow$$

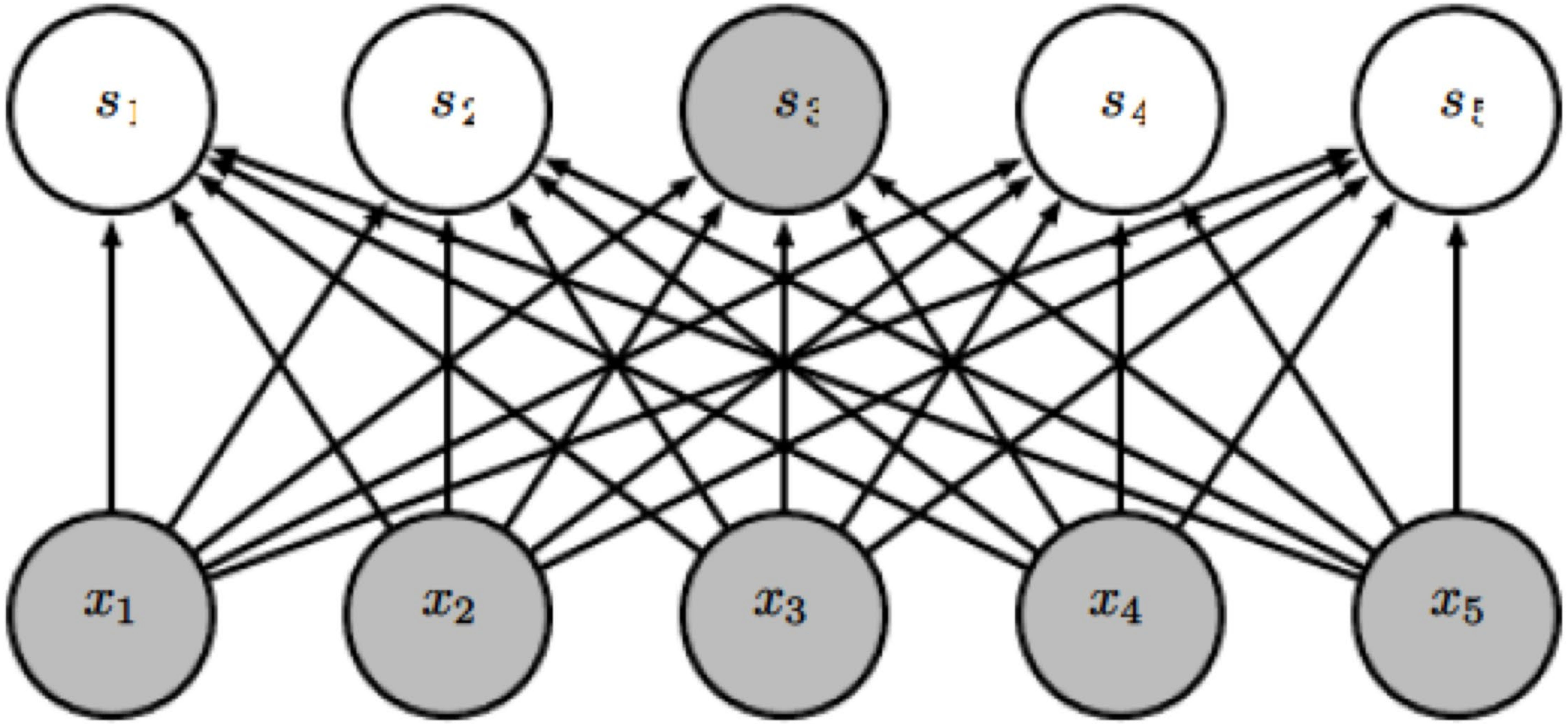
Gaussian Blur

Convolutional Neural Networks

- Convolutional networks: neural networks that use convolution in place of general matrix multiplication in at least one of their layers
- Strong empirical performance in applications – particularly computer vision.
- Examples: image classification, object detection.

Advantage: sparse interaction

Fully connected layer, $m \times n$ edges



m output nodes

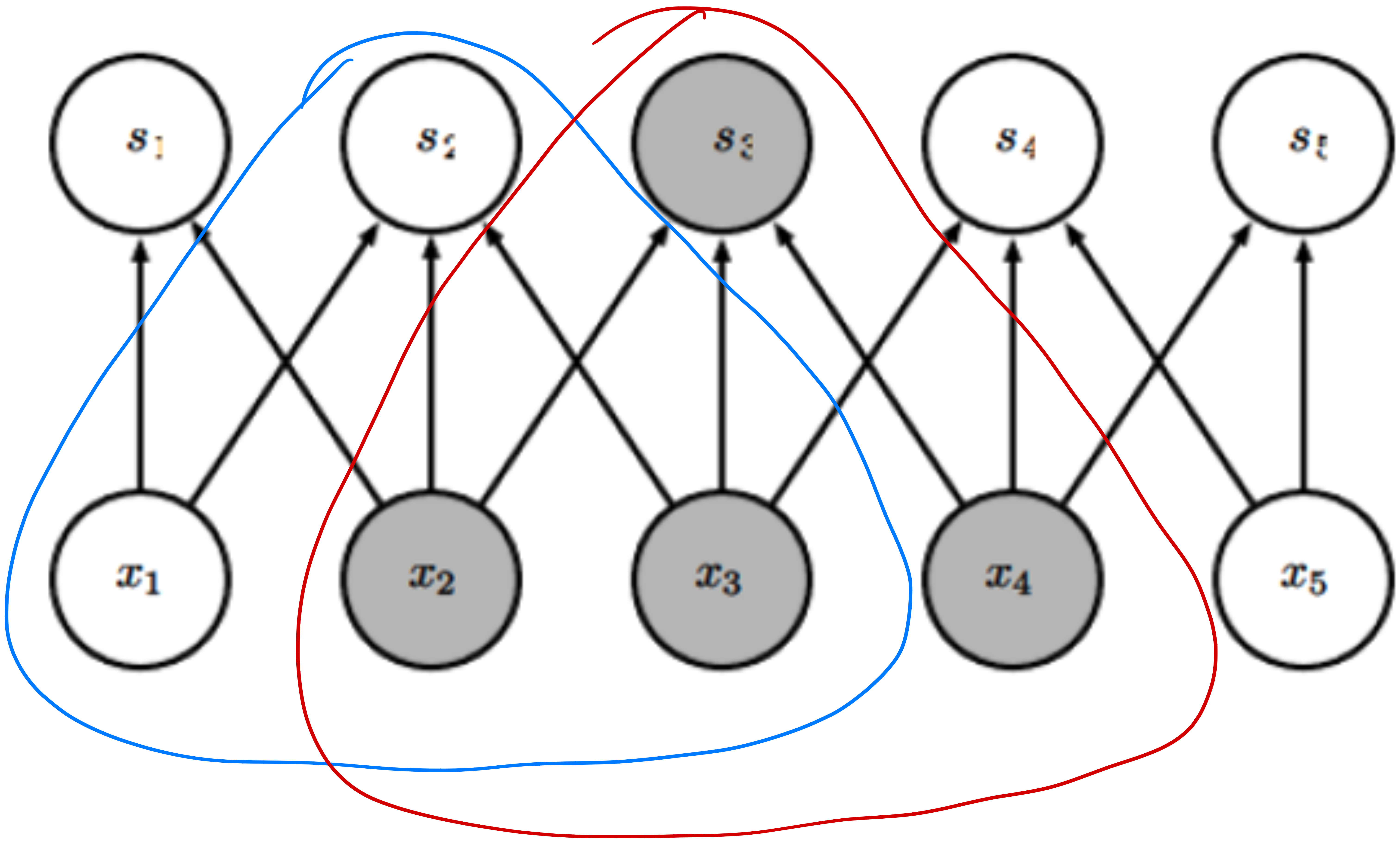
n input nodes

Figure from *Deep Learning,* by Goodfellow, Bengio, and Courville

Advantage: sparse interaction

★ Advantage: we are not connecting everything with the other row

Convolutional layer, $\leq m \times k$ edges



m output nodes

k kernel size

n input nodes

Figure from *Deep Learning,* by Goodfellow, Bengio, and Courville

Q1. Suppose we want to perform convolution as follows. What's the output?

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} + 1 = ?$$

A. $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 1 \\ 3 & 4 \end{bmatrix}$

(0+ 1+3-4) = 0 → +1 = 1 (Top-left)

(0 +2 +4-5) = 1 → +1 = 2 (Top-right)

(0 + 4 +6 -7) = 3 → +1 = 4 (Bottom-left)

Q1. Suppose we want to perform convolution as follows. What's the output?

$$\begin{bmatrix}0&1&2\\3&4&5\\6&7&8\end{bmatrix} * \begin{bmatrix}0&1\\1&-1\end{bmatrix} \boxed{+1} = \begin{bmatrix}1&2\\4&5\end{bmatrix}$$

$$0\times0+1\times1+3\times1+4\times(-1)+1=1$$
$$1\times0+2\times1+4\times1+5\times(-1)+1=2$$
$$3\times0+4\times1+6\times1+7\times(-1)+1=4$$
$$4\times0+5\times1+7\times1+8\times(-1)+1=5$$

A. $\begin{bmatrix}1&2\\4&5\end{bmatrix}$

B. $\begin{bmatrix}1&2\\3&4\end{bmatrix}$

$\begin{bmatrix}1&3\\3&5\end{bmatrix}$

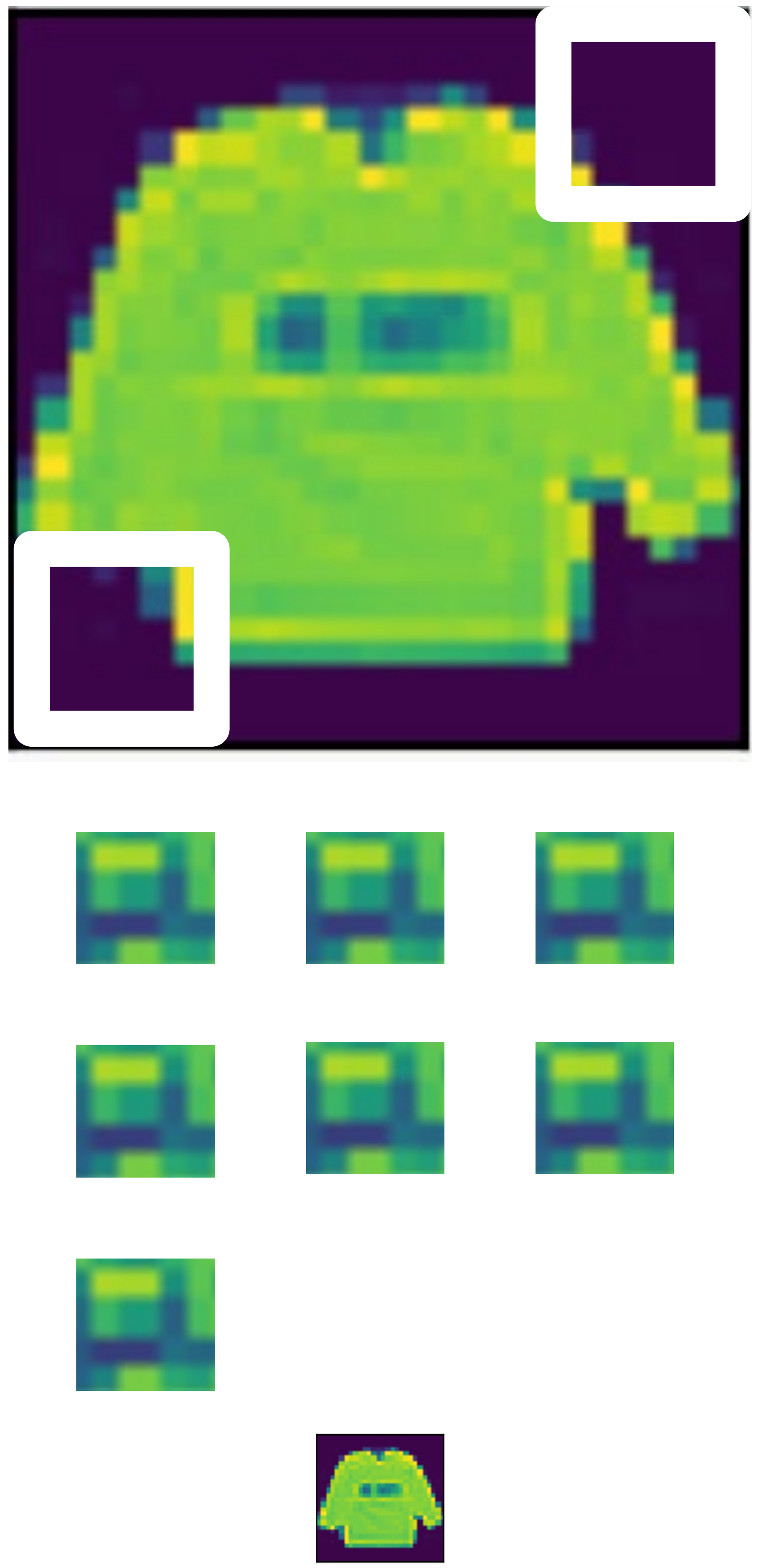
B. $\begin{bmatrix}0&1\\3&4\end{bmatrix}$

Padding and Stride

Padding

- Given a 32 x 32 input image
- Apply convolution with 5 x 5 kernel
  - 28 x 28 output with 1 layer
  - 4 x 4 output with 7 layers

(★ Recall that our output is $(M_h - K_h + 1) \times (M_w - K_w + 1)$, so if our kernel size is larger, it means our output size will be smaller)

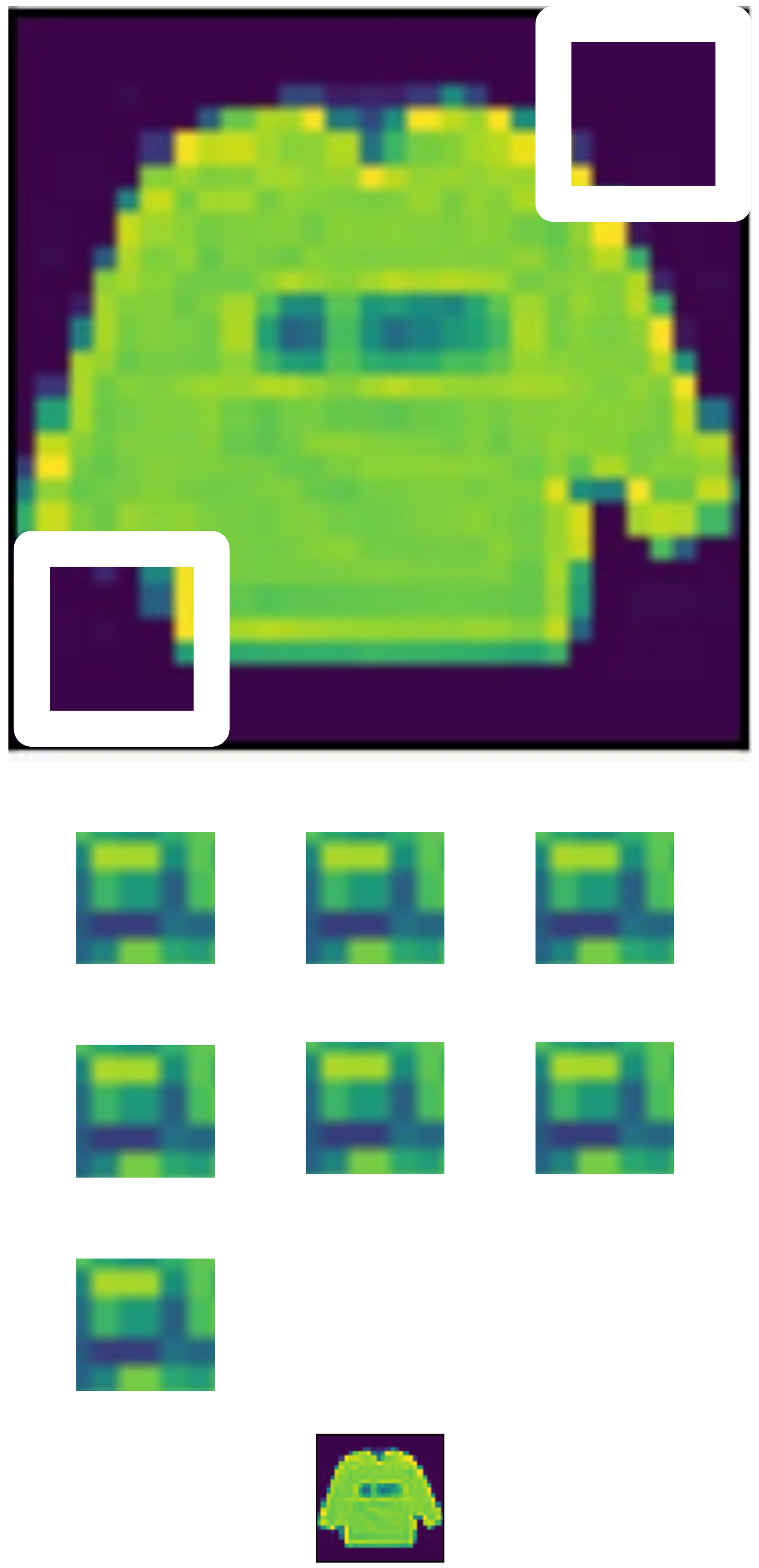


Padding

- Given a 32 x 32 input image
- Apply convolution with 5 x 5 kernel
  - 28 x 28 output with 1 layer
  - 4 x 4 output with 7 layers
- Shape decreases faster with larger kernels
  - Shape reduces from $n_h \times n_w$ to

$$(n_h - k_h + 1) \times (n_w - k_w + 1)$$

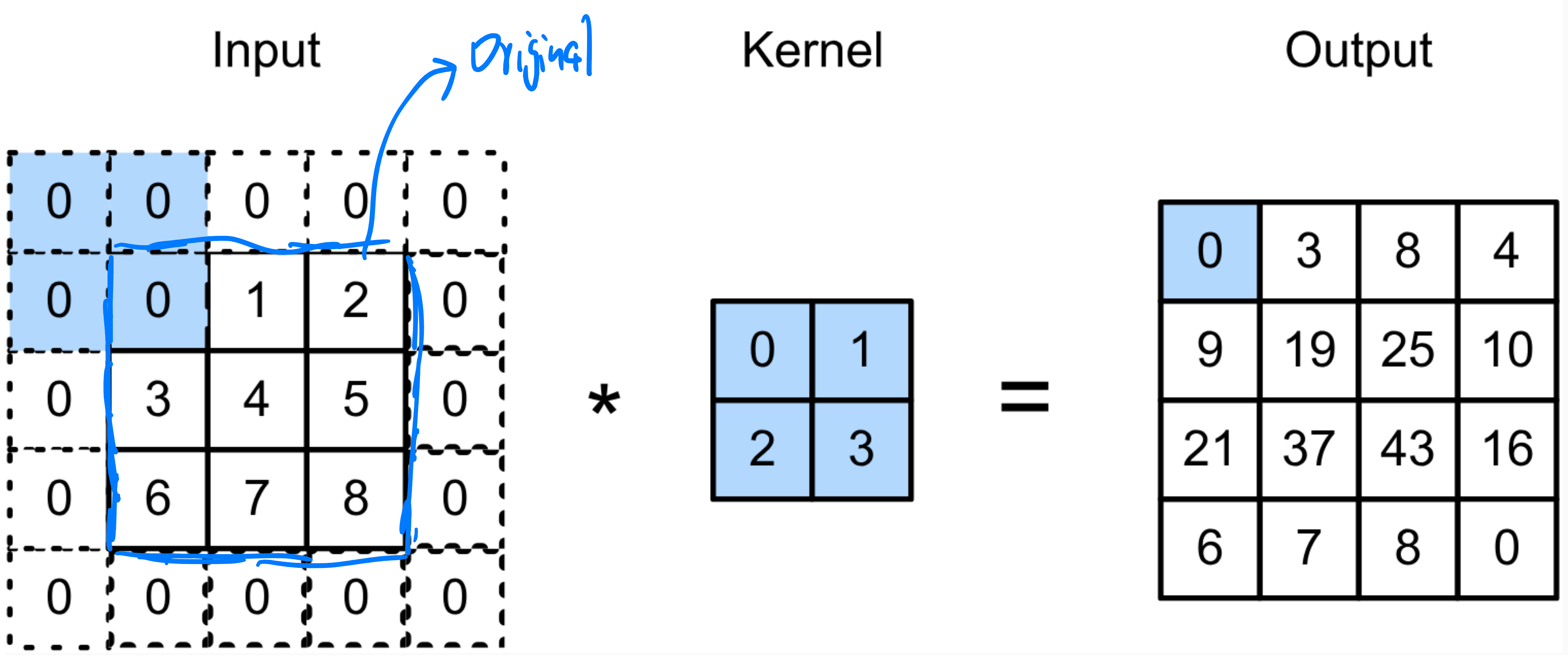
(★ And we want to avoid it, we don't want to lose too much information on the output)

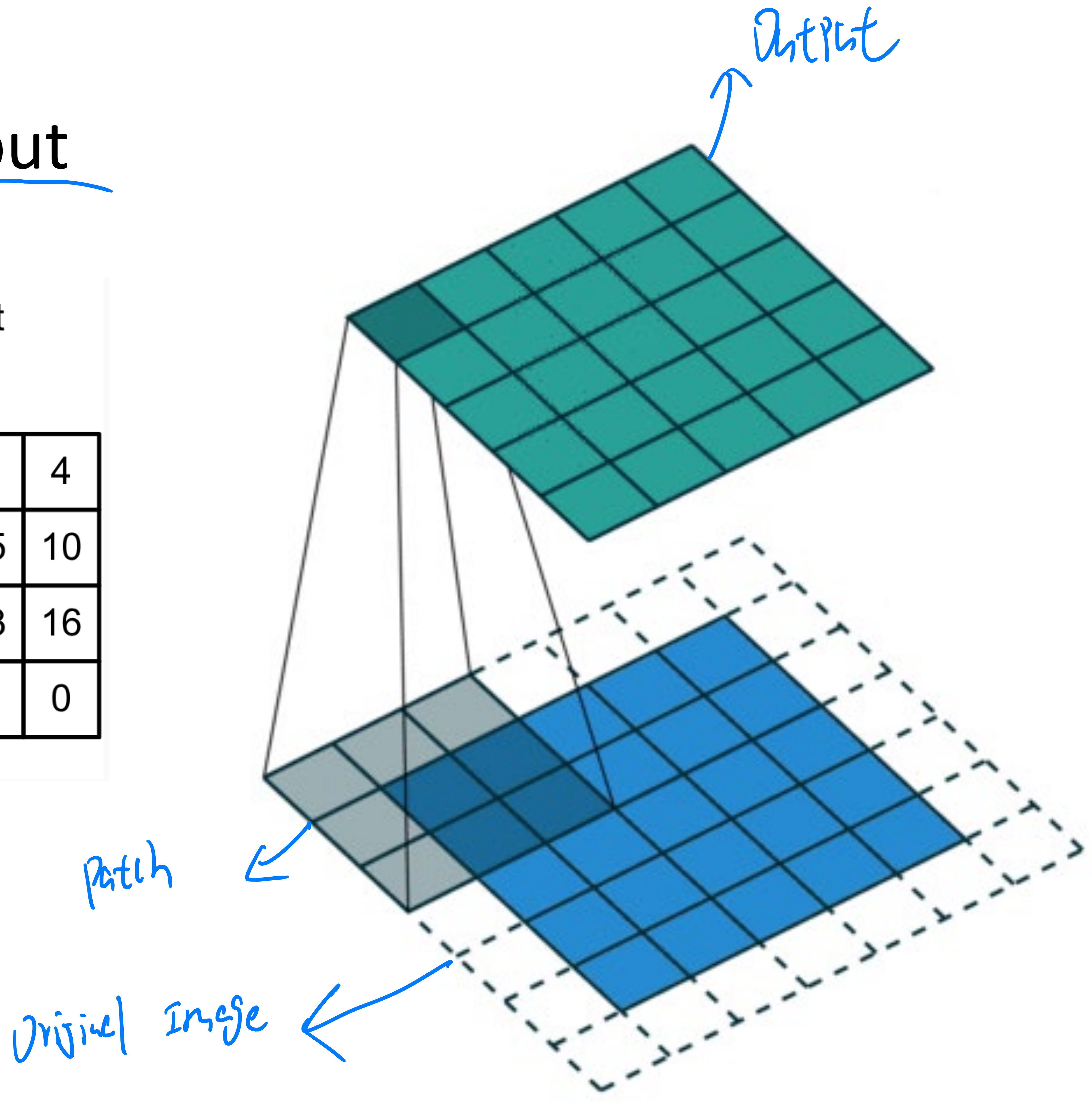


Convolutional Layers: Padding

more specifically, adding 0s.

★ **Padding** adds rows/columns around input



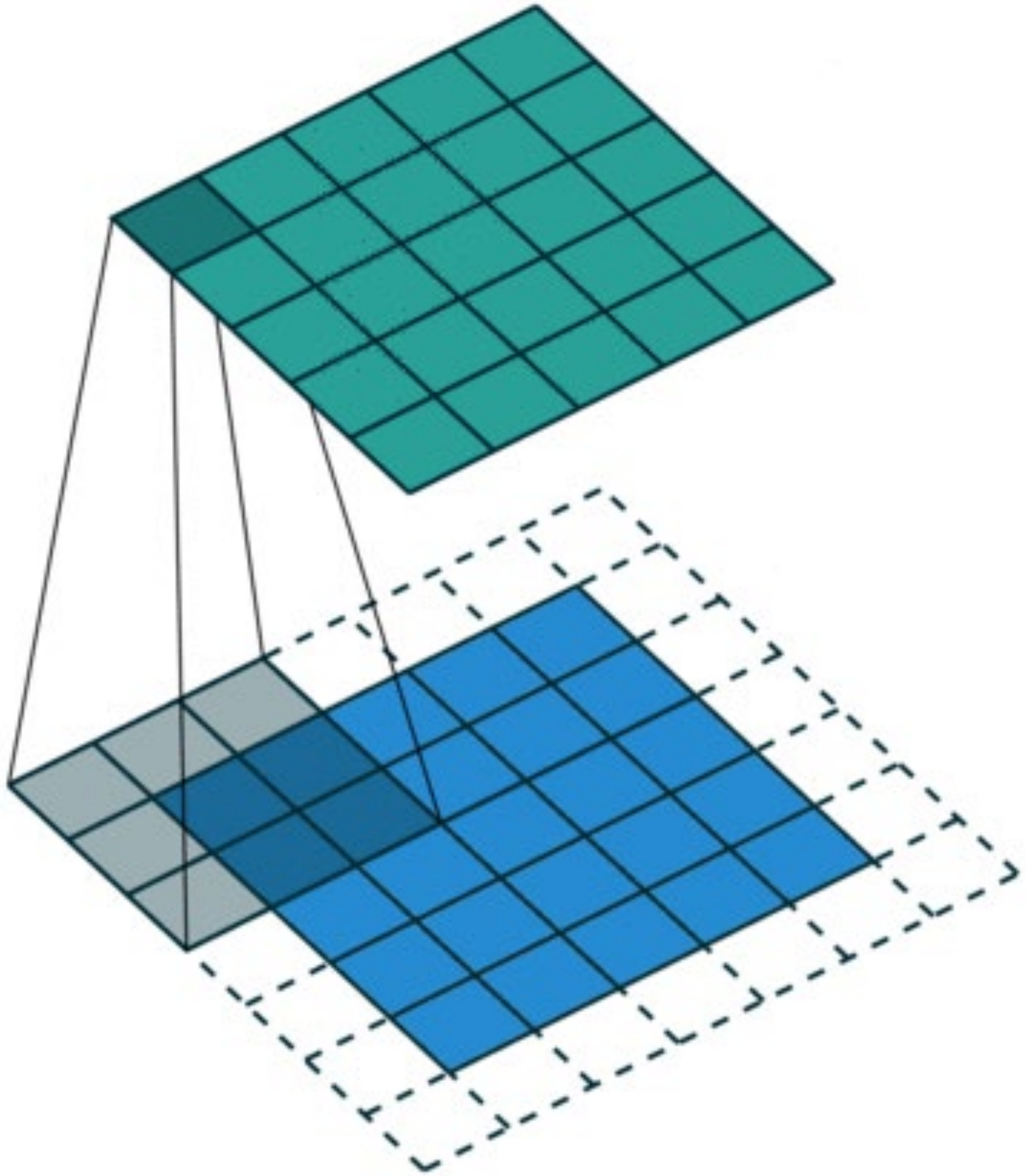


Convolutional Layers: Padding

Padding adds rows/columns around input

- Why?

1. Keeps **edge information**
2. Preserves sizes / allows deep networks
   - ie, for a 32x32 input image, 5x5 kernel, after 1 layer, get 28x28, after 7 layers, **only 4x4**
3. Can combine different filter sizes



Convolutional Layers: Padding

- Padding p_h rows and p_w columns, output shape is

$$(n_h\text{-}k_h\text{+}p_h\text{+}1) \text{ x } (n_w\text{-}k_w\text{+}p_w\text{+}1)$$

- Common choice is $p_h = k_h\text{-}1$ and $p_w\text{=}k_w\text{-}1$ → Input size = output size
  - Odd k_h: pad $p_h/2$ on both top and bottom
  - Even k_h: pad ceil($p_h/2$) on top, floor($p_h/2$) on bottom

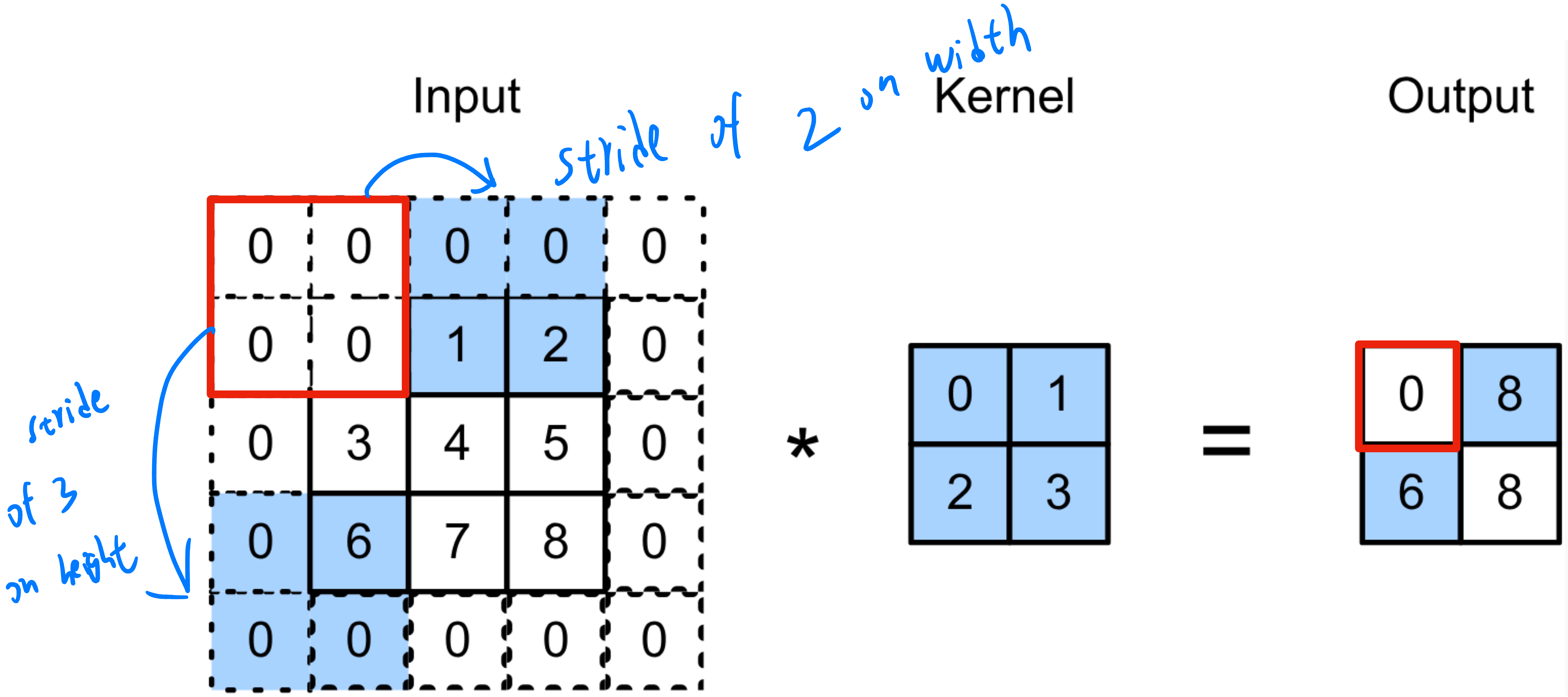
One more row on the top than the bottom

Stride

- Stride is the #rows / #columns per slide

Example: strides of 3 and 2 for height and width

★ (Previously, our slide is one at a time, the stride simply accelerating this process by taking many steps at a time for the filter)



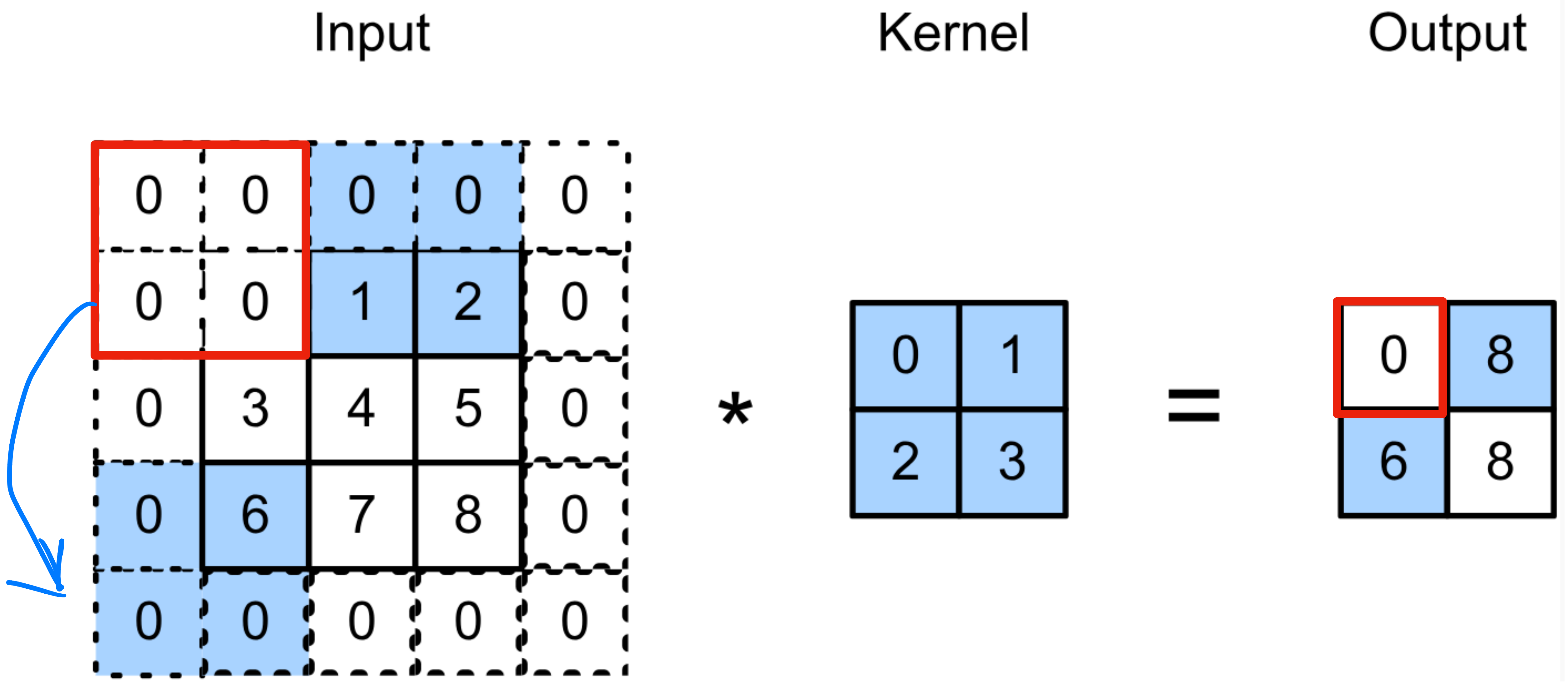
(★ And we often need both padding and the strides)

$$0\times0 + 0\times1 + 1\times2 + 2\times3 = 8$$
$$0\times0 + 6\times1 + 0\times2 + 0\times3 = 6$$

Stride

- Stride is the #rows / #columns per slide

Example: strides of 3 and 2 for height and width



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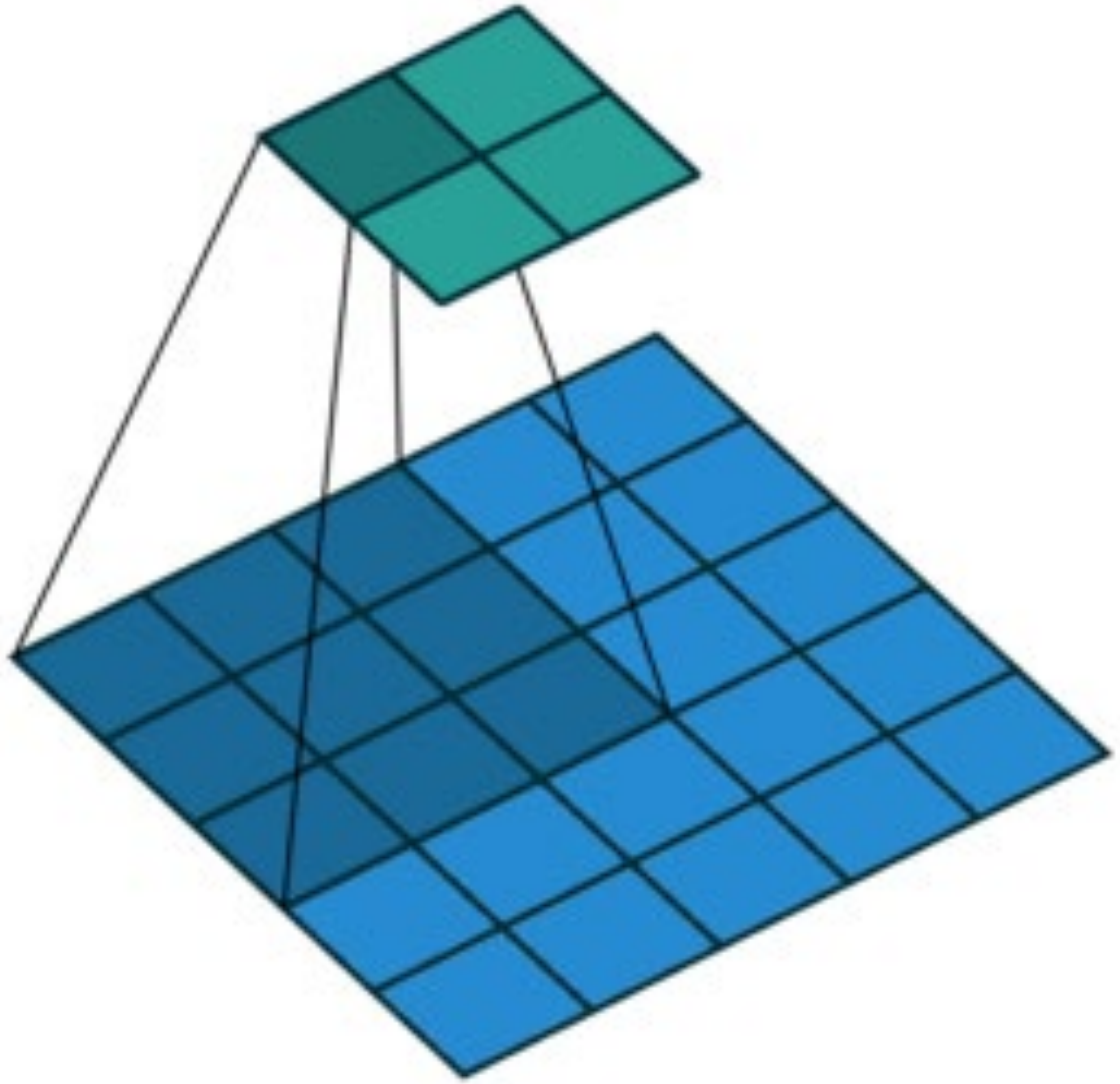


Stride 2,2

Convolutional Layers: Stride

- Given stride s_h for the height and stride s_w for the width, the output shape is

☆ Notice we are taking the floor now ✓

$$\lfloor(n_h-k_h+p_h+\underbrace{s_h)/s_h}_{\text{stride height}}\rfloor \times \lfloor(n_w-k_w+p_w+\underbrace{s_w)/s_w}_{\text{stride width}}\rfloor$$

- Set $p_h = k_h-1$, $p_w = k_w-1$, then get

$$\lfloor(n_h+s_h-1)/s_h\rfloor \times \lfloor(n_w+s_w-1)/s_w\rfloor$$

Q2. Suppose we want to perform convolution on a single channel image of size 7x7 (no padding) with a kernel of size 3x3, and stride = 2. What is the dimension of the output?

A. 3x3

B. 7x7

C. 5x5

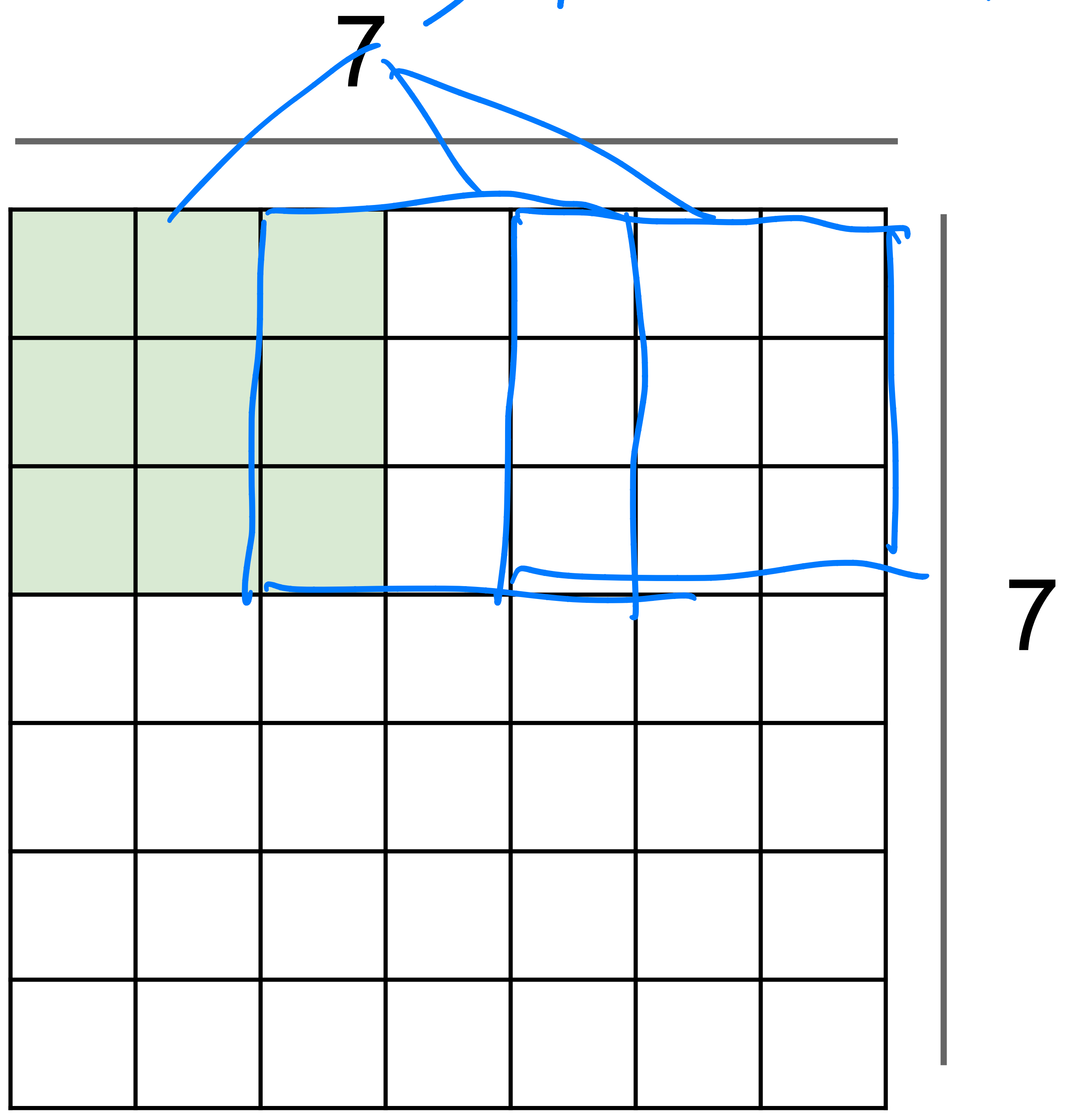
D. 2x2

$\lfloor (7-3+2)/2 \rfloor = 3$

$\therefore$ 3x3

3 horizontally

$\therefore$ 3x3 also



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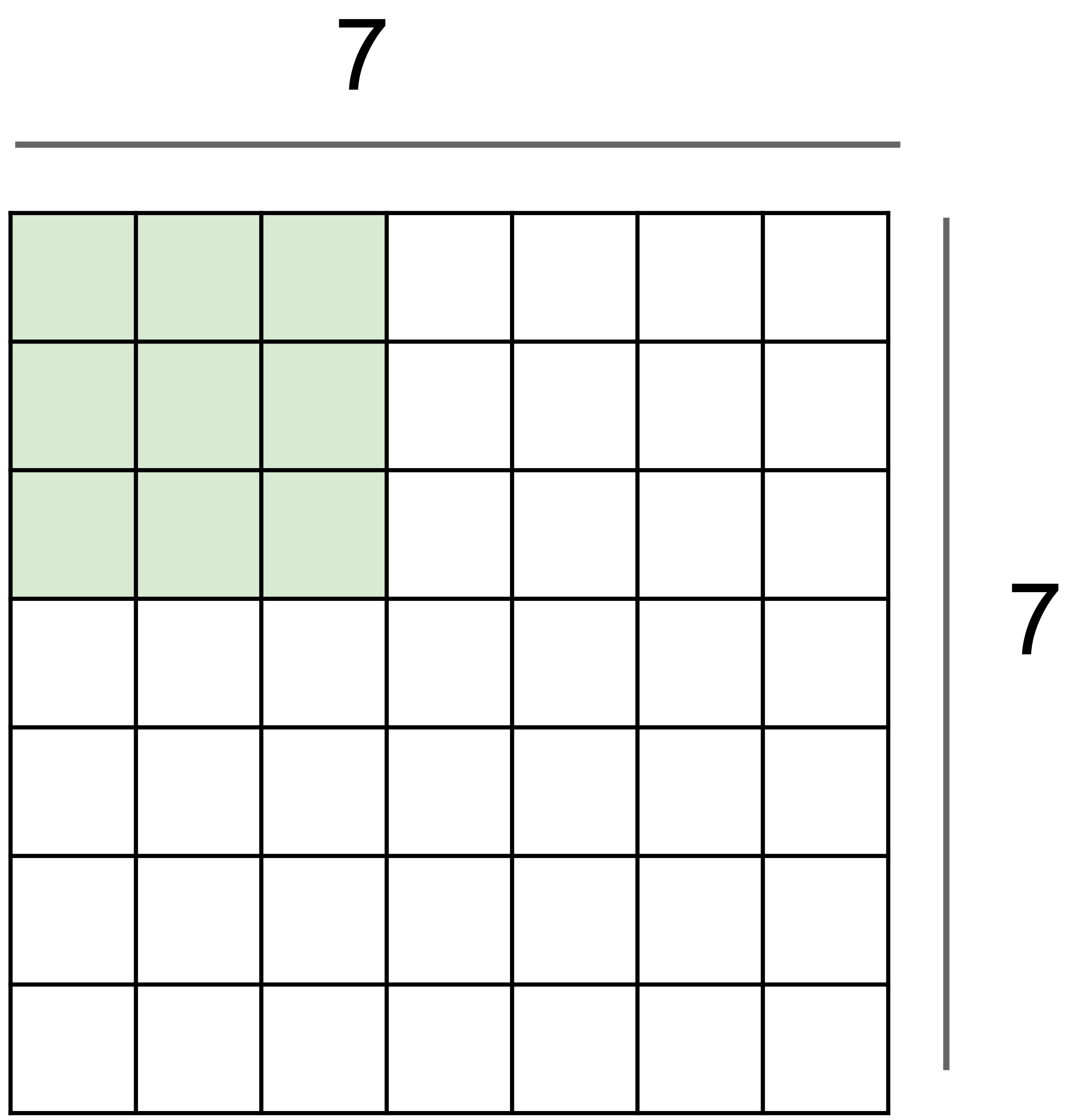
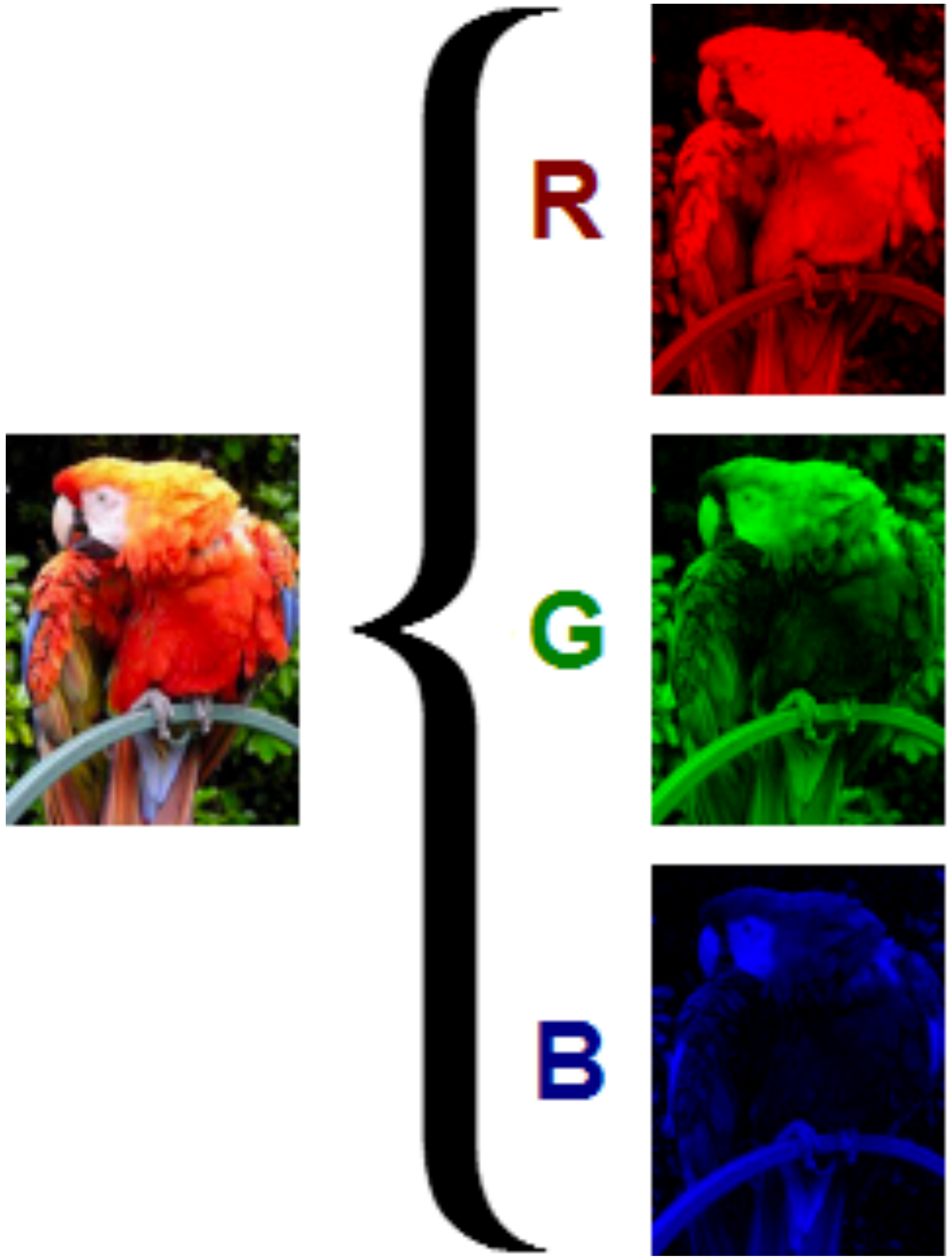


$$\lfloor(n_h-k_h+p_h+s_h)/s_h\rfloor \times \lfloor(n_w-k_w+p_w+s_w)/s_w\rfloor$$

Multiple Input and Output Channels

Multiple Input Channels

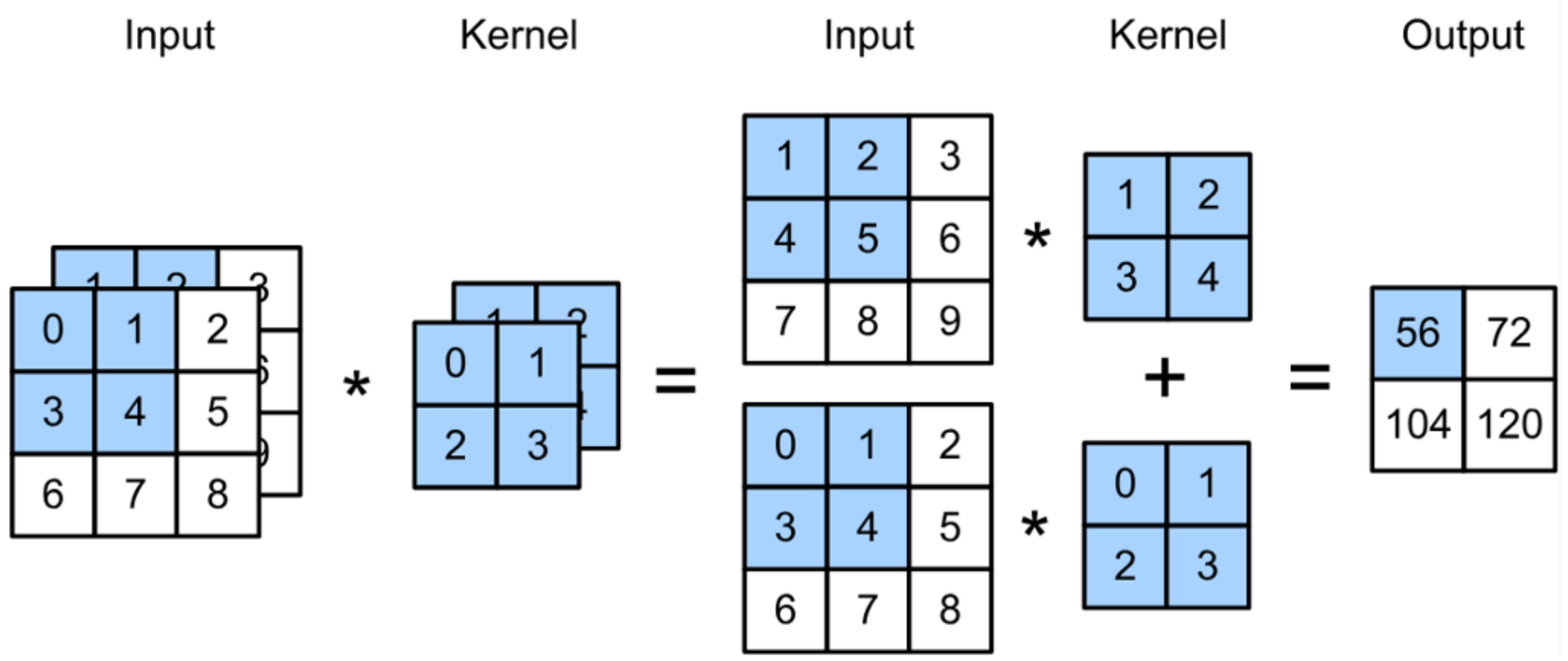
- Color image may have three RGB channels
- Converting to grayscale loses information



Multiple Input Channels

- Have a kernel matrix for each channel, and then sum results over channels

multiple input/kernel → sum all of them together to get the output



$$(1 \times 1 + 2 \times 2 + 4 \times 3 + 5 \times 4)$$
$$+(0 \times 0 + 1 \times 1 + 3 \times 2 + 4 \times 3)$$
$$= 56$$

Convolutional Layers: Channels

- How to integrate multiple channels?
  - Have a kernel for each channel, and then sum results over channels

input channel

$$\mathbf{X} : c_i \times n_h \times n_w$$

$$\mathbf{W} : c_i \times k_h \times k_w$$

width

height

$$\mathbf{Y} : m_h \times m_w$$

$$\mathbf{Y} = \sum_{i=0}^{c_i} \mathbf{X}_{i,:,:} \star \mathbf{W}_{i,:,:}$$

"Slices" of tensors

Tensor: generalization of matrix to higher dimensions

Multiple Output Channels

- No matter how many inputs channels, so far we always get single output channel
- We can have **multiple 3-D kernels**, each one generates an output channel

Multiple Output Channels

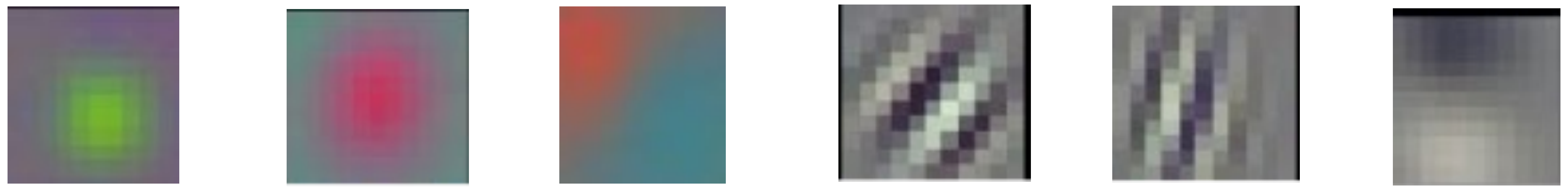
- No matter how many inputs channels, so far we always get single output channel
- We can have **multiple 3-D kernels**, each one generates an output channel
- Input $\mathbf{X} : c_i \times n_h \times n_w$
- Kernels $\mathbf{W} : c_o \times c_i \times k_h \times k_w$
- Output $\mathbf{Y} : c_o \times m_h \times m_w$

$$\mathbf{Y}_{i,:,:} = \mathbf{X} \star \mathbf{W}_{i,:,:,:}$$
$$\text{for } i = 1, \ldots, c_o$$

(☆ Why: so far, each kernel is doing one type of searching operation, but we'd like to solve any classification problems, which requires lots of kernel acting in parallel. So we just repeat this process multiple times with different kernels.)

Multiple Input/Output Channels

- Each 3-D kernel may recognize a particular pattern



(Gabor filters)

Q3. Suppose we want to perform convolution on an RGB image of size 224x224 (no padding) with 64 kernels, each with height 3 and width 3. Stride = 1. Which is a reasonable estimate of the total number of scalar multiplications involved in this operation (without considering any optimization in matrix multiplication)?

A. 64 x 3 x 3 x 222 x 222

B. 64 x 3 x 3 x 222

C. 3 x 3 x 222 x 222

D. 64 x 3 x 3 x 3 x 222 x 222

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B. 64 x 3 x 3 x 222

C. 3 x 3 x 222 x 222

D. 64 x 3 x 3 x 3 x 222 x 222

For each kernel, we slide the window to 222 x 222 different locations. For each location, the number of multiplication is 3x3x3. So in total 64x3x3x3x222x222

$\because$ RGB → 3 channels → 3x3x3

(222x222 entries of output → 222x222 times of multiplications on a single location, and that number is determined by the size of the kernel : 3x3)

Q4. Suppose we want to perform convolution on a RGB image of size 224 x 224 (no padding) with 64 kernels, each with height 3 and width 3. Stride = 1. The convolution layer has bias parameters. Which is a reasonable estimate of the total number of learnable parameters?

A. 64 x 222 x 222

B. 64 x 3 x 3 x 222

C. 3 x 3 x 3 x 64

D. (3 x 3 x 3 + 1) x 64

Q4. Suppose we want to perform convolution on a RGB image of size 224 x 224 (no padding) with 64 kernels, each with height 3 and width 3. Stride = 1. The convolution layer has bias parameters. Which is a reasonable estimate of the total number of learnable parameters?

A. 64 x 222 x 222

B. 64 x 3 x 3 x 222

C. 3 x 3 x 3 x 64

D. (3 x 3 x 3 + 1) x 64

Q4. Suppose we want to perform convolution on a RGB image of size 224 x 224 (no padding) with 64 kernels, each with height 3 and width 3. Stride = 1. The convolution layer has bias parameters. Which is a reasonable estimate of the total number of learnable parameters?

A. 64 x 222 x 222

B. 64 x 3 x 3 x 222

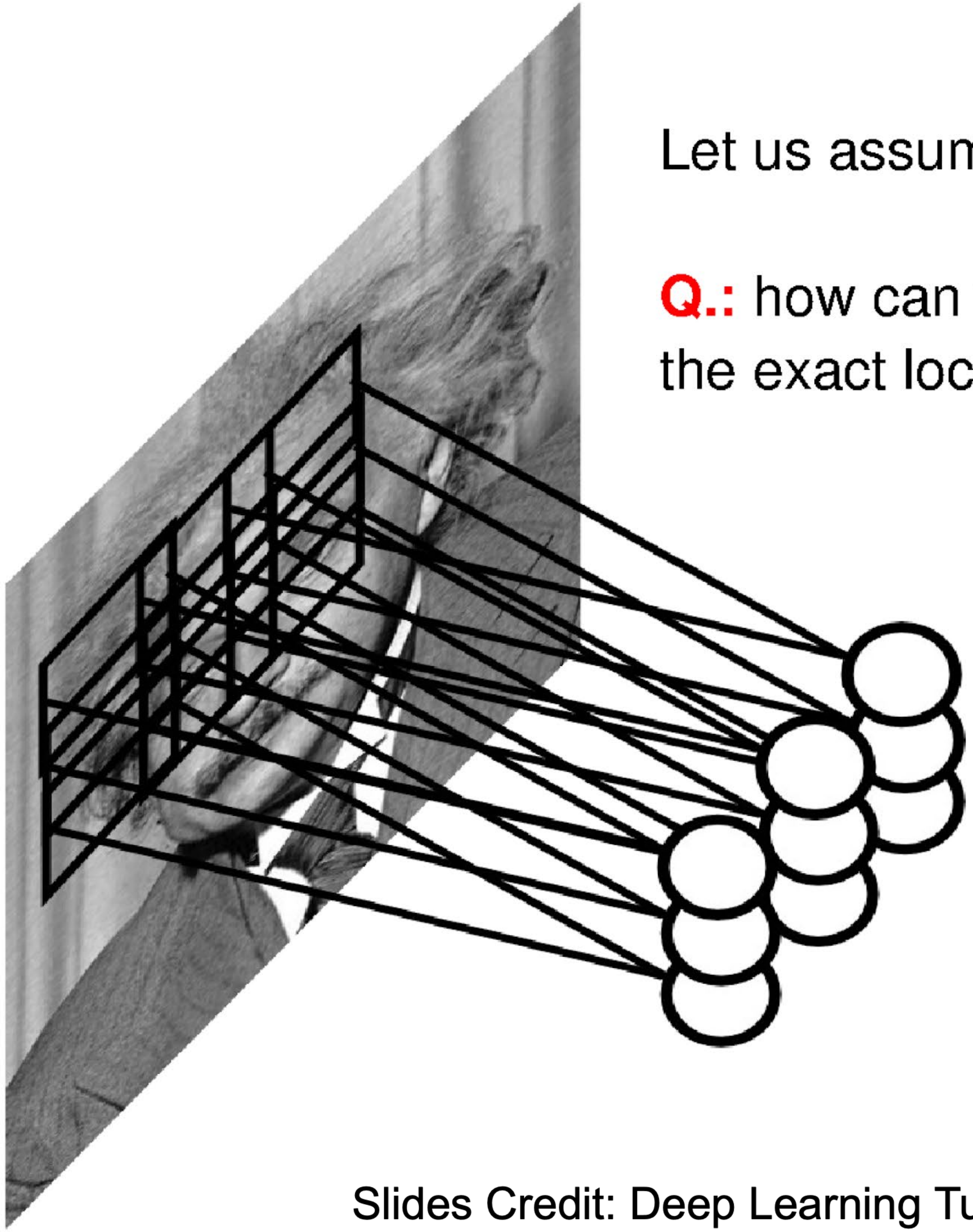
C. 3 x 3 x 3 x 64

D. (3 x 3 x 3 + 1) x 64

Each kernel is 3D kernel across 3 input channels, so has 3x3x3 parameters. Each kernel has 1 bias parameter. So in total (3x3x3+1)x64

Pooling Layer

Pooling

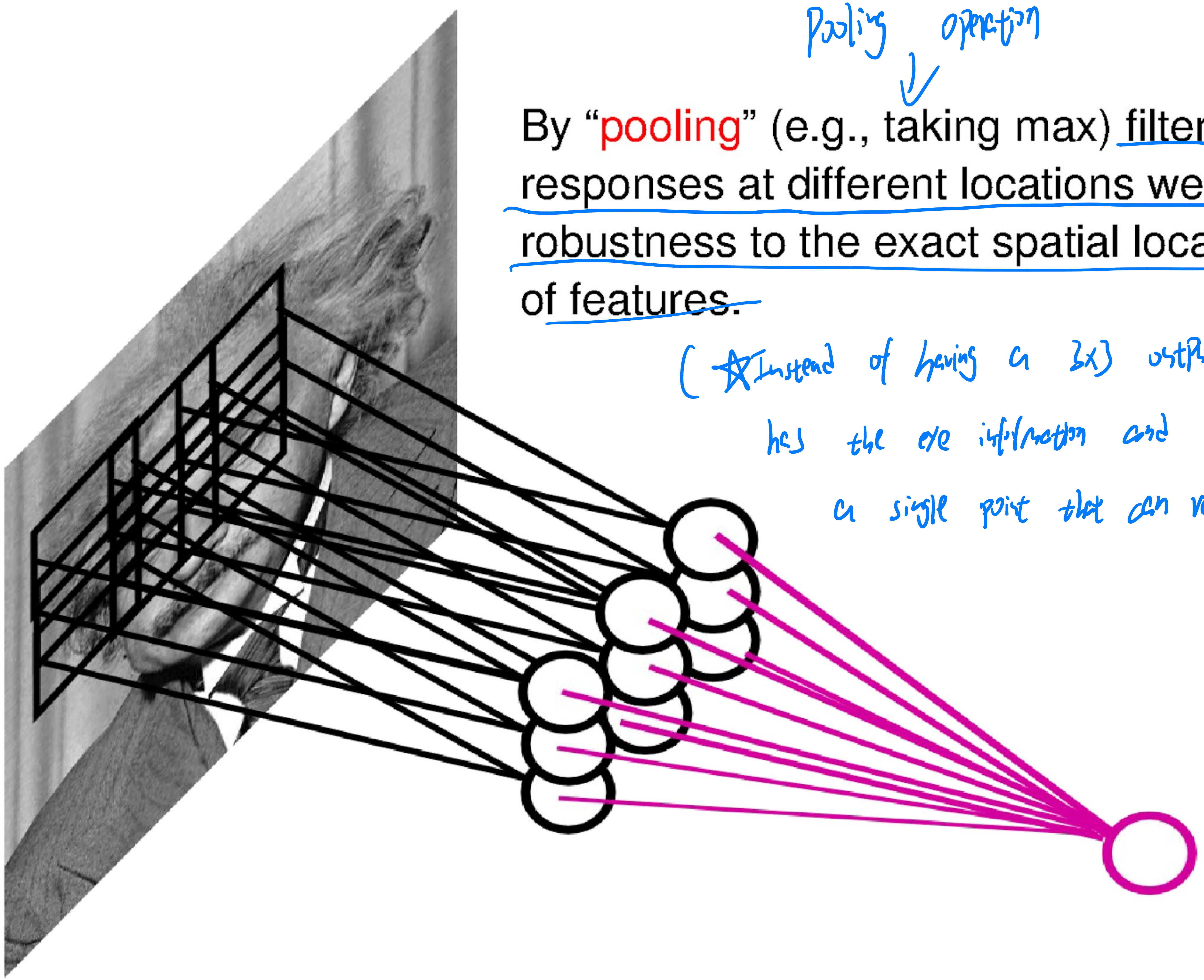


Let us assume filter is an “eye” detector.

Q.: how can we make the detection robust to the exact location of the eye?

Slides Credit: Deep Learning Tutorial by Marc’Aurelio Ranzato

Pooling



Pooling operation

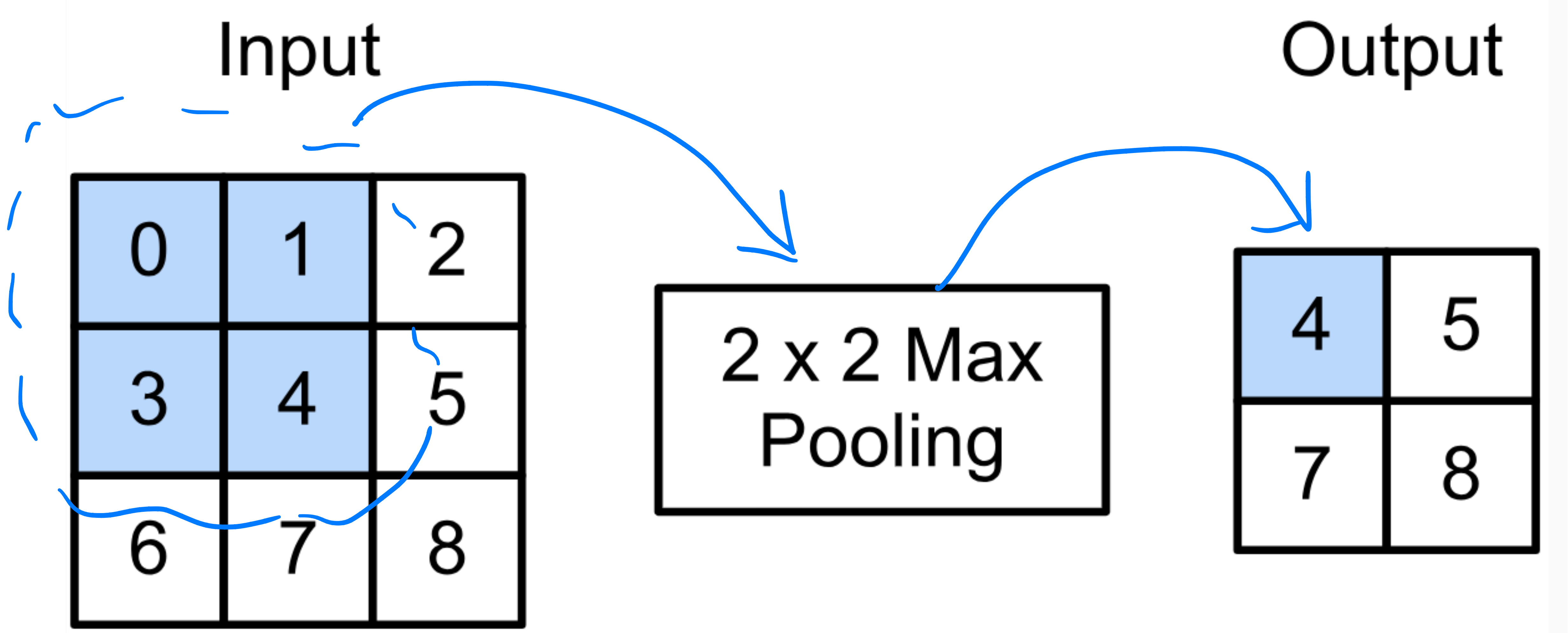
By "pooling" (e.g., taking max) filter responses at different locations we gain robustness to the exact spatial location of features.

(★Instead of having a 3x3 output telling me which location has the eye information and which does not, we want a single point that can represent yes/no (exact location))

Slides Credit: Deep Learning Tutorial by Marc'Aurelio Ranzato

2-D Max Pooling

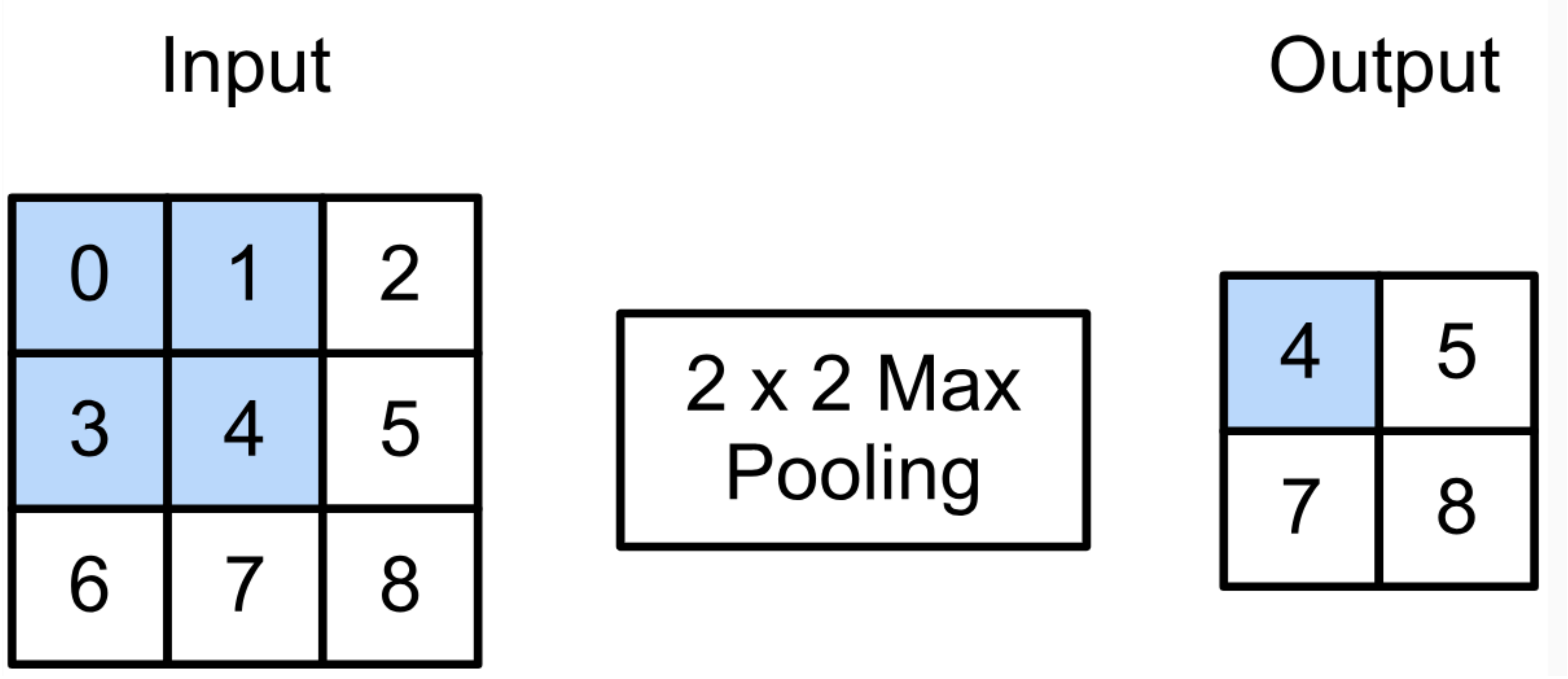
- Returns the maximal value in the sliding window



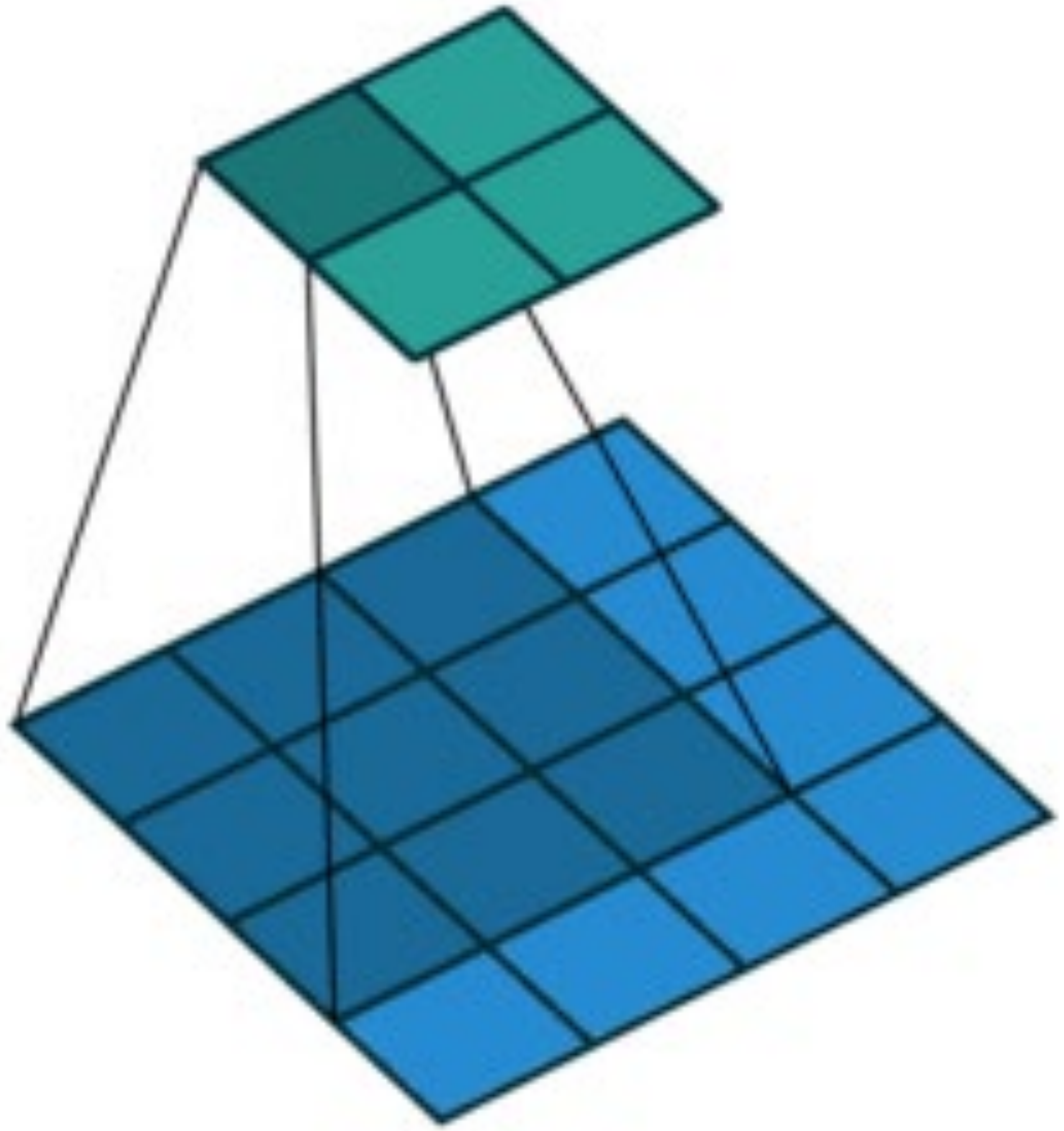
max(0,1,3,4) = 4

2-D Max Pooling

- Returns the maximal value in the sliding window



$\max(0,1,3,4) = 4$



Padding, Stride, and Multiple Channels

- Pooling layers have similar padding and stride as convolutional layers
- No learnable parameters
- Apply pooling for each input channel to obtain the corresponding output channel

#output channels = #input channels

(In pooling, each layer is just a fixed operation. Thus it is channel-wise, Each input is processed separately to produce a corresponding output channel. And if you have 3 in, then there will also be 3 out.)

Padding, Stride, and Multiple Channels

- Pooling layers have similar padding and stride as convolutional layers
- No learnable parameters
- Apply pooling for each input channel to obtain the corresponding output channel

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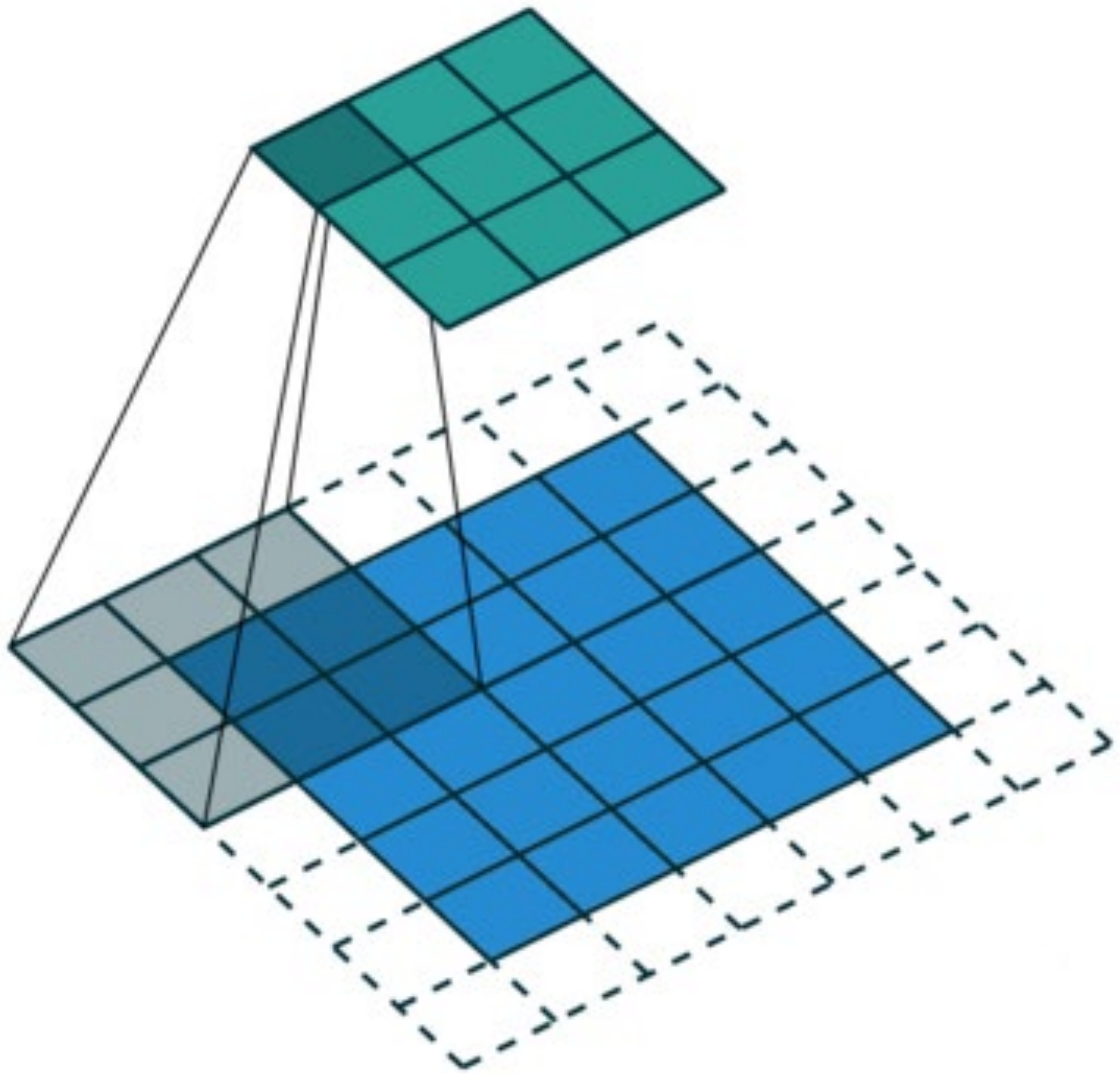
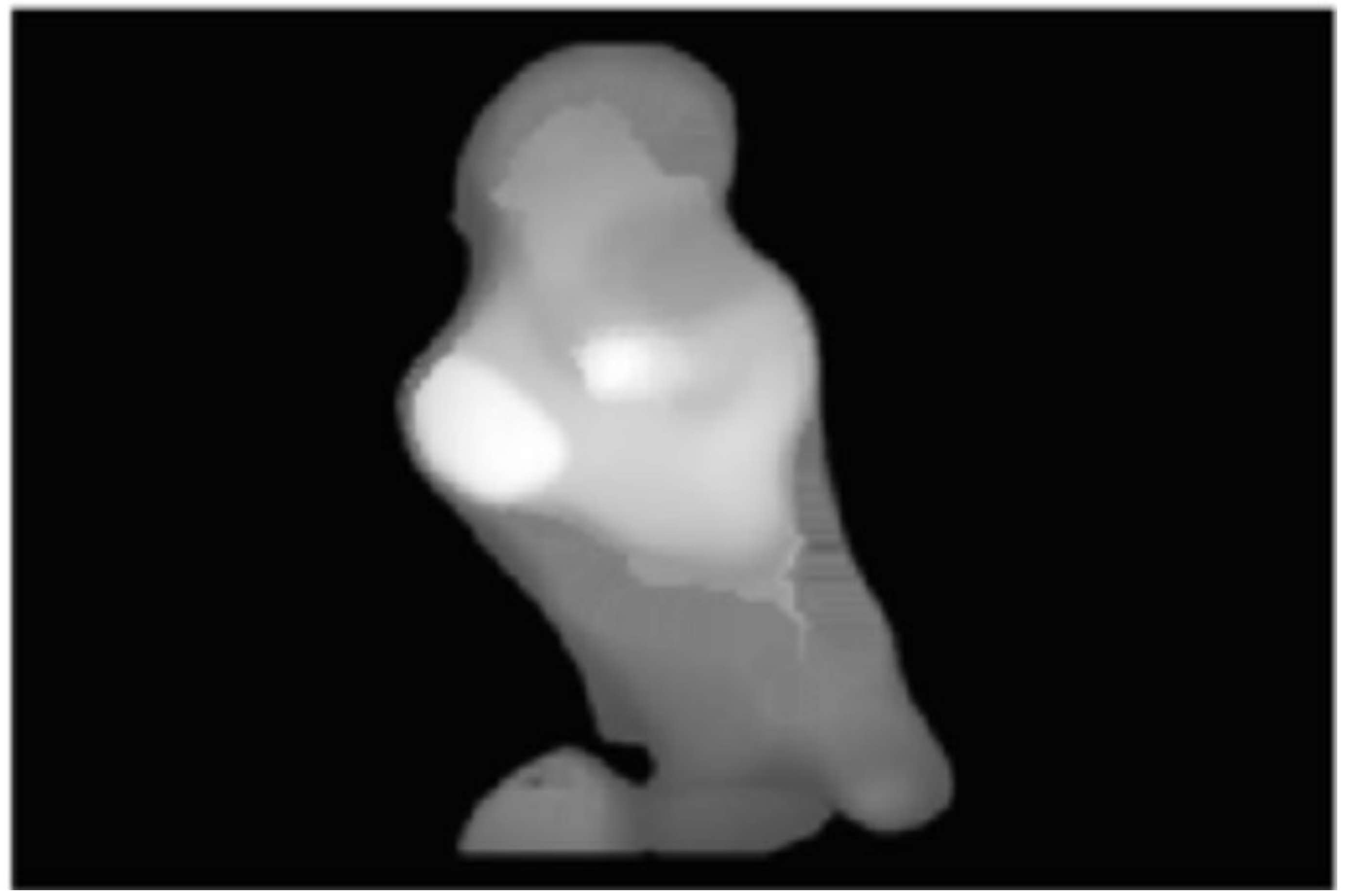


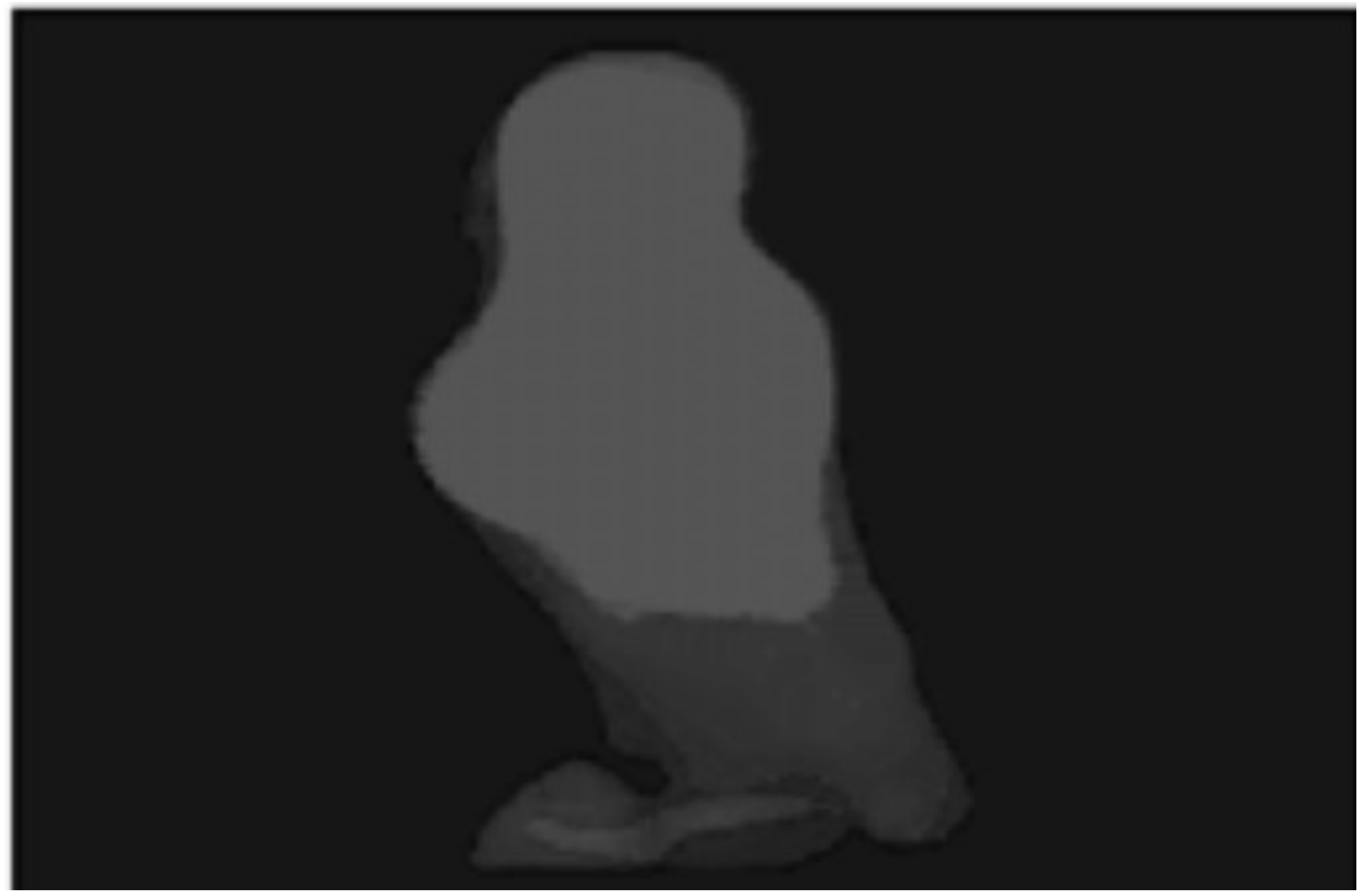
Average Pooling

- Max pooling: the strongest pattern signal in a window
- Average pooling: replace max with mean in max pooling
  - The average signal strength in a window

Max pooling



Average pooling



Q5. Suppose we want to perform 2x2 average pooling on the following single channel feature map of size 4x4 (no padding), and stride = 2. What is the output?

12	**20**	**30**	**0**
20	12	2	0
0	70	5	2
8	2	90	3

A.

20	**30**
70	90

B.

16	**8**
20	25

C.

20	**30**
20	25

D.

12	**2**
70	5

Q5. Suppose we want to perform 2x2 average pooling on the following single channel feature map of size 4x4 (no padding), and stride = 2. What is the output?

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20	**30**
20	25

D.

12	**2**
70	5

Q6. What is the output if we replace average pooling with 2 x 2 max pooling (other settings are the same)?

12	**20**	**30**	**0**
20	12	2	0
0	70	5	2
8	2	90	3

A.

20	**30**
70	90

B.

16	**8**
20	25

C.

20	**30**
20	25

D.

12	**2**
70	5

Q6. What is the output if we replace average pooling with 2 x 2 max pooling (other settings are the same)?

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20	12	2	0
0	70	5	2
8	2	90	3

A.

20	**30**
70	90

B.

16	**8**
20	25

C.

20	**30**
20	25

D.

12	**2**
70	5

Summary

- Intro of convolutional computations
    - 2D convolution
    - Padding, stride
    - Multiple input and output channels
    - Pooling

Acknowledgement:

Some of the slides in these lectures have been adapted from materials developed by Alex Smola and Mu Li:

https://courses.d2l.ai/berkeley-stat-157/index.html